Predicting Opioid Overdoses in Pennsylvania Using Neural Networks

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Abstract

Opioid-induced overdoses have resulted in the deaths of over 450,000 people in the United States. The surge in opioid addiction and opioid-induced deaths in the past two decades has developed into the largest drug crisis in American history. Despite this, attempts at data-driven forecasting of opioid overdose risk have been limited in the literature. This paper hypothesizes that local rates of opioid overdose incidents and deaths can be forecast using machine learning models. This hypothesis is evaluated on data from the state of Pennsylvania. Specifically, this paper uses neural networks for two different prediction tasks. A Long Short-Term Memory (LSTM) recurrent neural network is used for time-series forecasting of administrations of the anti-overdose drug Naloxone. A feedforward neural network is developed to perform county-level regression on opioid-induced death rate. These models and their predictions can be expanded to other geographies and can inform policy decisions among government and first responder personnel, hopefully resulting in policy that reduces deadly overdoses.

Background

Opioid medications are widely used as pain relievers. However, many prescription opioids carry a high risk of dependency and addiction. Additionally, abuse of common prescription opioids like Oxycodone can transition into abuse of far deadlier drugs such as heroin or the synthetic opioid fentanyl. Over the past two decades there has been a substantial rise in levels of opioid abuse and opioid-induced overdose death in the United States. The current opioid crisis is the largest and most deadly drug epidemic in this nation's history, with over 450,000 overdose deaths in this century.

Despite the scale of this crisis, there has been limited work about predicting overdose risk using machine learning or other computational models. Most existing research (Lo-Ciganic et al. 2019, Ellis et al. 2019) has focused on predicting the risk of abuse for patients already taking prescription opioids. Additional work has been done in data mining on social media (Sarker et al, 2019) to locate regions with high levels of abuse. However, this paper targets the broader task of overdose prediction for local regions. Models have been developed for this general geographic forecasting task (Wakeland, Nielsen, and Geissert, 2015), but they frequently rely on handcrafted simulations of underlying mechanisms (system dynamics). Correctly modeling all contributing factors can prove prohibitive in regions with limited data or for researchers with limited knowledge of mechanisms such as the price of illegal street drugs. Consequently, this paper proposes machine learning approaches to the opioid prediction task that enable useful prediction for a greater number of opioid-impacted populations.

Approach

This paper presents two distinct prediction tasks which lend themselves to different neural network architectures. The first task involves predicting the opioid-induced death rate in a county based on known demographic, socioeconomic, and temporal risk factors. This is developed into a feedforward neural network for regression. The model incorporates county-level features from the U.S. Census Bureau's American Community Survey selected through linear regression.

The second task involves predicting the number of daily administrations of Naloxone for a region. Naloxone is a powerful anti-overdose drug carried by first responders that can be administered to an overdosing patient. This time-series forecasting task lends itself to the use of recurrent neural networks, specifically a Long Short-Term Memory (LSTM) model. LSTM networks (Hochreiter and Schmidhuber, 1997) can learn both the long-term and short-run (weekly) trends observed in time-series analysis of opioid overdoses.

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The state of Pennsylvania provides an excellent testing ground for neural network-based overdose forecasting. The PA government maintains an extensive online portal of opioid-related databases (PA Opioid Data Dashboard) that provide a rich source for predictive data.

Data

The data used in this paper was aggregated from state and federal government repositories. It includes county-level opioid death rates (CDC) and county-level demographic data (Census) for the regression model. Data for the LSTM consists of administrations of Naloxone by first responders (Opioid Data Dashboard).

CDC WONDER

The Centers for Disease Control and Prevention maintains a database of rates of underlying and multiple causes for deaths. Death rates per 100,000 residents were queried for Pennsylvania counties using MCD - ICD-10 Codes: T40.0 (Opium); T40.1 (Heroin); T40.2 (Other opioids); T40.3 (Methadone); T40.4 (Other synthetic narcotics); T40.6 (Other and unspecified narcotics).

Census Bureau Demographic Data

The United States Census Bureau estimates a number of statistics annually using their American Community Survey. 1-year estimates of income, poverty, and employment data for the 67 counties in Pennsylvania were queried for the available years (2010-2018).

PA Opioid Data Dashboard

The Pennsylvania Department of Health aggregates all reported administrations of the anti-overdose drug Naloxone by emergency medical services (EMS) starting from Q1 2018. This paper transformed the ~12,000 discrete incidents into daily counts for modeling purposes.

Method

This paper proposes two models for the two proposed prediction tasks. The first is a regression model predicting the number of fatal opioid overdoses per 100,000 residents of a county. Using demographic features selected through linear regression, it employs a feedforward neural network to predict death rates for a given year. Selected features include population, year, unemployment rate, and labor force participation.

The regression model is a simple feed-forward dense neural network implemented in Tensorflow/Keras. It has two hidden layers, each with 64 neurons, and produces a single output corresponding to a county's opioid-induced death rate. The model is trained on the CDC and Census data. It uses RMSProp with learning rate = 0.001 to minimize mean squared error. It trains with early stopping based on validation set loss.

The second model employs a LSTM recurrent neural network also implemented with Tensorflow/Keras. This sequential model uses a single LSTM layer followed by a dense layer to output predicted daily counts of Naloxone administrations in the state. It trains using the Adam optimizer to minimize mean average error and performs early stopping.

Results

Evaluating the true predictive accuracy of the developed models is impossible before the release of new overdose data from the Pennsylvania state government and the CDC. However, model performance on test sets not seen during training provide an indication of their utility. Models were evaluated using the metrics feature of the scikit-learn Python module on hold-out test sets comprising 20% of total data. The regression model was able to explain 47% of the variance in its test data. Such performance is encouraging given the limitations in data size. The LSTM model showed less promising performance, explaining only 17% of variance in its test set. Despite its larger initial training set, this model was limited by data compression through transformation into daily counts, along with the greater difficulty of its prediction task.

Conclusion

The results in this paper provide a promising first step toward more widespread use of machine learning models for general opioid overdose prediction. The methods described can easily be applied to other geographies and can be adapted to suit the needs of smaller monitoring regions.

This work could be improved by incorporating more high-quality data. Municipality-level reporting from local EMS companies in particular would enable more features and greater prediction precision for both model types. However, such detailed reporting also carries privacy, security, and bias concerns.

This research is aimed at many constituent groups, including first responders, local/county governments, state officials, other researchers, and residents of opioid-afflicted areas. First responders and officials may use research predictions to influence policy decisions, while researchers can implement predictive models for other states or experiment with new model architectures. Finally, residents of Pennsylvania can use this research to be more informed and prepared for the impact of opioids on their community.

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